Shreyas Devdatta Khobragade

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Professional Summary

Robotics Engineer with an **M.S. in Robotics (GPA 4.0)** and **2+ years** of hands-on experience designing and integrating full-stack **autonomous systems**, from real-time **perception** to closed-loop **motion planning** and **control**. Skilled in **ROS2, C, C++, Python, and PyTorch** for **sensor fusion**, **3D reconstruction**, **deep learning** model optimization, and **sim-to-real deployment** on resource-constrained **embedded platforms (Jetson Orin, Raspberry Pi)**. **First-author** publications at **IEEE RA-L 2026** and **ICRA Workshop**, with experience collaborating with PIs from prototype through manuscript submission.

Education

Worcester Polytechnic Institute — **Worcester, MA**
M.S., Robotics and Automation Engineering   GPA: 4.0 / 4.0 — *Aug 2024 – May 2026*
- **Awards:** Dr. Glenn Yee Graduate Student Project Award, Travel Award, EDU Bridge Scholarship

Visvesvaraya National Institute of Technology — **Nagpur, India**
B.Tech., Electronics and Communication Engineering   GPA: 8.26 / 10 — *Aug 2018 – May 2022*

Work and Research Experience

Perception and Autonomous Robotics (PeAR) Lab, WPI — **Worcester, MA**
Graduate Researcher – Aerial Autonomy, Perception, and Sim-to-Real Deployment Project — *Jul 2025 – May 2026*
- Designed and implemented a **sim-to-real** autonomy pipeline for aerial robots in zero-light, GPS-denied environments, published at **IEEE RA-L 2026**.
- Built a **depth estimation** system using structured lighting and coded apertures for obstacle avoidance in complete darkness ($<$**1 milli-lux illumination**).
- Trained and deployed a **DenseNet** model using **PyTorch** & **TensorRT**, reaching **20 Hz** real-time inference on **Jetson Orin Nano** under latency constraints.
- Integrated perception with motion planning in a closed-loop controller, achieving a **95.5%** autonomous navigation success rate in cluttered environments.
- Fused **event camera** and structured-light sensing to extend obstacle detection to 6 meters in complete darkness, including thin and dark objects.
- Boosted perception throughput to **30+ fps** on **Jetson Orin Nano** by reimplementing the event-based pipeline in **C++**, enabling **7 m/s** flight.
- Engineered two custom quadrotors end-to-end, spanning mechanical design, electronics integration, and flight testing, validating both perception systems.

Jio Platforms Limited — **Bengaluru, India**
5G Software Engineer, R&D — *Jun 2022 – May 2024*
- Developed and optimized high-performance **C/C++** networking software in the Vector Packet Processing (VPP) framework for production 5G core systems.
- Implemented **shared-memory** instrumentation for UPF statistics by modifying VPP source in C, enabling faster crash debugging and monitoring.
- Migrated and refactored UPF components across newer VPP releases, improving system stability, performance, and maintainability.

Publications

- **S. Khobragade***, D. Singh*, N. J. Sanket, “AsterNav: Autonomous Aerial Robot Navigation in Darkness Using Passive Computation,” *IEEE Robotics and Automation Letters (RA-L)*, 2026. Paper Link
- **S. Khobragade***, B. Vaghasiya*, D. Singh*, N. J. Sanket, “NightSight: Passive Computation for Navigation in Dark Using Events,” *Neuromorphic Field Robotics Workshop, IEEE ICRA*, 2026. Paper Link
- *(Under Review)* D. Singh, **S. Khobragade***, B. Vaghasiya*, M. Bhat, N. J. Sanket, “Dhruva: Aerial Navigation Using Coded Events,” *Nature Communications*, 2026.

Technical Projects

Structure-from-Motion & NeRF Multi-View 3D Reconstruction | Github — Jan 2025 – Mar 2025
- Engineered a classical **Structure-from-Motion (SfM)** pipeline incorporating **SIFT** feature matching, **RANSAC** 8-point fundamental matrix estimation, non-linear **PnP** pose refinement ($\sim$16% error reduction), and sparse bundle adjustment using SciPy's least-squares optimizer for global consistency.
- Trained an 8-layer **NeRF** model with positional encoding, reaching PSNR 28.4 on the Lego benchmark, and built a custom capture pipeline (**COLMAP**, Polycam, EXIF calibration, background removal) reaching PSNR 31.49 / SSIM 0.94 on self-collected physical figurine data.

Real-Time Motion Planning for Drones in Unknown Environments | Github — Oct 2024 – Dec 2024
- Developed a real-time **C++** motion planning framework integrating **OctoMap** for hierarchical global 3D occupancy mapping from simulated **LiDAR point clouds** across cluttered environments, and **EDT3D** for local collision detection via distance transforms within a small windowed region around the drone.
- Designed a kinodynamic **RRT**-based planner in **OMPL** generating safe, efficient trajectories under vehicle dynamics, validated in **Gazebo** simulation using **PX4** SITL bridged to **ROS** via MAVROS for closed-loop, collision-free navigation testing.

Einstein Vision: Advanced Perception Stack for Self-Driving Cars | Github — Jan 2025 – May 2025
- Integrated **object detection** and **tracking (YOLO, Detic)**, instance segmentation (Mask R-CNN), lane detection, monocular depth **(Depth Anything V2)**, and **RAFT optical flow** into a unified perception pipeline on Tesla Model S footage rendered in Blender.
- Generated high-fidelity rendered videos across 13 driving sequences, cutting perception ambiguity 15–20% via classical heuristics.

Learning Visual Inertial Odometry for Aerial Robots | Github — Jan 2025 – May 2025
- Implemented a stereo MSCKF **VIO** pipeline in C++/Python featuring IMU state propagation, feature tracking, and EKF updates with chi-squared gating, achieving 0.091 m translation RMSE on the EuRoC benchmark for robust state estimation.
- Developed a CNN+LSTM sensor fusion network fusing visual and IMU features, reaching 2.38m RMSE across 25 different trajectories.

Technical Skills

- **Perception, Computer Vision, and 3D:** Real-time Perception Pipelines, 3D Reconstruction (NeRF, Structure-from-Motion, Point Clouds, Photogrammetry), Camera Calibration and Pose Estimation, Depth Estimation (Depth-from-Defocus, Monocular), Object Detection and Tracking, Object Segmentation (YOLOv8, Detic), Optical Flow (RAFT), Event Cameras, Visual-Inertial Odometry (VIO), OpenCV, Multi-View Geometry
- **Autonomy, Planning, and Control:** Motion and Path Planning (OMPL, RRT, OctoMap, Nav2), Manipulator Kinematics (Forward/Inverse Kinematics, DH Parameters), MoveIt, PD/PID Control (Motor Control), Closed-Loop Autonomous Navigation, Sim-to-Real Deployment, Reinforcement Learning (PPO), GPS-Denied Navigation, SLAM, EKF State Estimation, VLM, VLA, Sensor Integration and Sensor Fusion (RGB-D/RealSense, LiDAR, IMU)
- **Software and Infrastructure:** Python, C++, C, Embedded C, MATLAB, ROS, ROS2, Linux, Docker, Git/GitHub, PyTorch, TensorFlow, TensorRT, Unit Testing, CI/CD, 32-bit Microcontrollers, RTOS, Fusion 360, Onshape, SolidWorks, protocols: CAN, SPI, I2C, UART
- **Robotics Stack and Simulation:** PX4 Autopilot, ArduPilot, Mavlink, Isaac Sim, Gazebo, MuJoCo, RViz, Simulink, Blender
- **Hardware:** Jetson Orin Nano/NX, Raspberry Pi, Intel RealSense, Pixhawk Flight Controller, Arduino Uno, ESP32